

VIT CHENNAI · OFFICIAL SYLLABUS ALIGNED

Engineering Mathematics Complete Study Guide

Concept explanations, highlighted formulas, and solved examples for all core math courses. Built for VIT Chennai students — module by module, exactly as taught.

BMAT101L

Calculus

BMAT102L

Diff. Equations & Transforms

BMAT201L

Complex Variables & Linear
Algebra

BMAT202L

Probability & Statistics

VIT ENGINEERING MATHEMATICS SERIES

Calculus

Single Variable · Multivariable · Multiple Integrals · Special Functions · Vector Calculus

Module 1: Single Variable Calculus

Rolle's Theorem and Mean Value Theorem

KEY THEOREMS

Rolle's Theorem: If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$

Mean Value Theorem (Lagrange): $f'(c) = \frac{f(b) - f(a)}{b - a}$ for some $c \in (a, b)$

Maxima and Minima (Single Variable)

FIRST AND SECOND DERIVATIVE TESTS

Critical point: $f'(x) = 0$ or $f'(x)$ undefined

Second derivative test: If $f'(c) = 0$ and $f''(c) > 0 \rightarrow$ local minimum; $f''(c) < 0 \rightarrow$ local maximum

Concave up: $f''(x) > 0$; **Concave down:** $f''(x) < 0$

Inflection point: where $f''(x) = 0$ and concavity changes

— SOLVED EXAMPLE — MAXIMA/MINIMA

Find the maxima and minima of $f(x) = x^3 - 3x^2 - 9x + 5$

- 1 Find critical points: $f'(x) = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1) = 0$, so $x = 3, -1$
- 2 $f''(x) = 6x - 6$
- 3 At $x = -1$: $f''(-1) = -12 < 0 \rightarrow$ **local maximum**. $f(-1) = -1 - 3 + 9 + 5 = 10$
- 4 At $x = 3$: $f''(3) = 12 > 0 \rightarrow$ **local minimum**. $f(3) = 27 - 27 - 27 + 5 = -22$

Local max = 10 at $x = -1$; Local min = -22 at $x = 3$

Integration — Area and Volumes

KEY FORMULAS

Area between curves: $A = \int_a^b [f(x) - g(x)] dx$ where $f(x) \geq g(x)$

Volume (disc method): $V = \pi \int_a^b [f(x)]^2 dx$

Volume (shell method): $V = 2\pi \int_a^b x f(x) dx$

Module 2–3: Multivariable Calculus

Partial Derivatives and Total Differential

KEY DEFINITIONS

Partial derivative $\partial f / \partial x$: Differentiate w.r.t. x , treating y as constant

Total differential: $df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$

Jacobian: $J\left(\frac{u, v}{x, y}\right) = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{vmatrix}$

Chain rule: $\frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt}$

Taylor's Series for Two Variables

FORMULA

$$f(a + h, b + k) = f(a, b) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right) f + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}\right)^2 f + \dots$$

Maxima and Minima of $f(x, y)$

CONDITIONS (SECOND ORDER TEST)

Find critical point: $f_x = 0$ and $f_y = 0$

Compute discriminant: $D = f_{xx}f_{yy} - (f_{xy})^2$

If $D > 0$ and $f_{xx} > 0 \rightarrow$ **minimum**

If $D > 0$ and $f_{xx} < 0 \rightarrow$ **maximum**

If $D < 0 \rightarrow$ **saddle point**

If $D = 0 \rightarrow$ **test fails**

Lagrange's Multiplier Method

To find extremum of $f(x, y, z)$ subject to constraint $g(x, y, z) = 0$:

METHOD

Form: $\nabla f = \lambda \nabla g$, i.e. $f_x = \lambda g_x$, $f_y = \lambda g_y$, $f_z = \lambda g_z$

Solve these equations simultaneously with $g = 0$ to find the constrained extremum

— SOLVED EXAMPLE — LAGRANGE MULTIPLIERS

Find the maximum and minimum of $f = x^2 + y^2$ subject to $x + y = 1$

- 1 Set up: $\nabla f = \lambda \nabla g \rightarrow 2x = \lambda, 2y = \lambda \rightarrow x = y$
- 2 Substitute into constraint: $x + x = 1 \rightarrow x = y = \frac{1}{2}$

Minimum value $= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ at $(1/2, 1/2)$

Module 4: Multiple Integrals

KEY FORMULAS

Double integral (Cartesian): $\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$

Change to polar: $x = r \cos \theta, y = r \sin \theta, dA = r dr d\theta$

Change of order: Sketch the region, re-identify the limits

Triple integral: $\iiint_V f(x, y, z) dV$

Cylindrical coords: $x = r \cos \theta, y = r \sin \theta, z = z, dV = r dr d\theta dz$

Spherical coords: $x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi, dV = \rho^2 \sin \phi d\rho d\phi d\theta$

Module 5: Special Functions

BETA AND GAMMA FUNCTIONS

$$\Gamma(n) = \int_0^\infty e^{-t} t^{n-1} dt, \quad \Gamma(n+1) = n \Gamma(n), \quad \Gamma(1/2) = \sqrt{\pi}$$

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\text{Useful reduction: } \int_0^{\pi/2} \sin^m \theta \cos^n \theta d\theta = \frac{\Gamma(\frac{m+1}{2}) \Gamma(\frac{n+1}{2})}{2 \Gamma(\frac{m+n+2}{2})}$$

$$\text{Error function: } \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt, \quad \operatorname{erfc}(x) = 1 - \operatorname{erf}(x)$$

Modules 6–7: Vector Calculus

Vector Differentiation

GRADIENT, DIVERGENCE, CURL

Gradient: $\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \rightarrow$ points in direction of maximum increase

Directional derivative: $D_{\hat{u}} f = \nabla f \cdot \hat{u}$

Divergence: $\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$

Curl: $\nabla \times \mathbf{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{vmatrix}$

Irrotational: $\nabla \times \mathbf{F} = \mathbf{0}$ **Solenoidal:** $\nabla \cdot \mathbf{F} = 0$

Vector Integration Theorems

THE THREE MAJOR THEOREMS

Green's Theorem (2D): $\oint_C (P dx + Q dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$

Stokes' Theorem: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$

Gauss Divergence Theorem: $\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{F} dV$

Exam tip — Green's Theorem

Often used to evaluate line integrals by converting to double integrals (or vice versa). Always check if the curve is closed and oriented counter-clockwise.

— SOLVED EXAMPLE — GAUSS DIVERGENCE THEOREM

Evaluate $\oiint_S \mathbf{F} \cdot \hat{n} dS$ where $\mathbf{F} = x^2 \hat{i} + y^2 \hat{j} + z^2 \hat{k}$ over the unit cube

- 1 Find divergence: $\nabla \cdot \mathbf{F} = 2x + 2y + 2z$
- 2 Apply Gauss: $\iiint_V (2x + 2y + 2z) dV = \int_0^1 \int_0^1 \int_0^1 (2x + 2y + 2z) dx dy dz$
- 3 By symmetry each integral gives the same value: $= 3 \times \int_0^1 \int_0^1 \int_0^1 2x dx dy dz = 3 \times 1 = 3$

Answer = 3

BMAT101L — Topics to Master for FAT

- ☐ Rolle's theorem, MVT — conditions and applications
- ☐ Maxima/minima of single and two-variable functions
- ☐ Lagrange multiplier problems
- ☐ Change of order and change of variables in double integrals
- ☐ Beta-Gamma function evaluations
- ☐ Gradient, divergence, curl — irrotational and solenoidal check
- ☐ Green's, Stokes', Gauss theorems — know when to apply each

Differential Equations and Transforms

ODE · PDE · Laplace · Fourier Series · Fourier Transform · Z-Transform

Module 1: Ordinary Differential Equations

2nd Order ODE with Constant Coefficients

GENERAL SOLUTION: $AY'' + BY' + CY = F(X)$

Solve auxiliary equation $am^2 + bm + c = 0$ for roots m_1, m_2

If $m_1 \neq m_2$ (real): $y_c = C_1 e^{m_1 x} + C_2 e^{m_2 x}$

If $m_1 = m_2 = m$ (repeated): $y_c = (C_1 + C_2 x) e^{mx}$

If $m = \alpha \pm i\beta$ (complex): $y_c = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$

PARTICULAR INTEGRAL — METHOD OF UNDETERMINED COEFFICIENTS

f(x)	Trial Particular Integral
e^{ax}	Ae^{ax} (if a is not a root)
$\sin ax$ or $\cos ax$	$A \sin ax + B \cos ax$
x^n	$A_n x^n + A_{n-1} x^{n-1} + \dots + A_0$
$e^{ax} \sin bx$	$e^{ax} (A \sin bx + B \cos bx)$

VARIATION OF PARAMETERS FORMULA

$$y_p = y_1 \int \frac{-y_2 f(x)}{W} dx + y_2 \int \frac{y_1 f(x)}{W} dx$$

where $W = y_1 y_2' - y_2 y_1'$ is the Wronskian

— SOLVED — 2ND ORDER ODE

Solve $y'' - 3y' + 2y = e^{3x}$

1 AE: $m^2 - 3m + 2 = 0 \rightarrow (m - 1)(m - 2) = 0 \rightarrow m = 1, 2$. So $y_c = C_1e^x + C_2e^{2x}$

2 PI: Try $y_p = Ae^{3x}$. Then $9Ae^{3x} - 9Ae^{3x} + 2Ae^{3x} = e^{3x} \rightarrow 2A = 1 \rightarrow A = \frac{1}{2}$

$$y = C_1e^x + C_2e^{2x} + \frac{1}{2}e^{3x}$$

Module 2: Partial Differential Equations

FORMATION AND STANDARD TYPES

Lagrange's PDE: $Pp + Qq = R$ where $p = \partial z / \partial x$, $q = \partial z / \partial y$. Solve using auxiliary equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

Type 1: $f(p, q) = 0$ — assume $z = ax + by + c$ where $f(a, b) = 0$

Type 2: $f(z, p, q) = 0$ — assume $z = f(x + ay)$

Separation of variables: Assume $u(x, t) = X(x) \cdot T(t)$, separate, solve two ODEs

Module 3–4: Laplace Transform

DEFINITION AND PROPERTIES

Definition: $\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt$

Linearity: $\mathcal{L}\{af + bg\} = aF(s) + bG(s)$

First shift: $\mathcal{L}\{e^{at} f(t)\} = F(s - a)$

Differentiation: $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$; $\mathcal{L}\{f''(t)\} = s^2 F(s) - sf(0) - f'(0)$

Integration: $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}$

Convolution: $\mathcal{L}\{f * g\} = F(s) \cdot G(s)$

STANDARD LAPLACE TRANSFORM TABLE

$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$
1	$1/s$
t^n	$n!/s^{n+1}$
e^{at}	$1/(s - a)$
$\sin at$	$a/(s^2 + a^2)$
$\cos at$	$s/(s^2 + a^2)$
$\sinh at$	$a/(s^2 - a^2)$
$\cosh at$	$s/(s^2 - a^2)$
$t \sin at$	$2as/(s^2 + a^2)^2$
$u(t - a)$	e^{-as}/s (Heaviside)
$\delta(t - a)$	e^{-as} (Dirac delta)

— SOLVED — SOLVE ODE USING LAPLACE

Solve $y'' + 4y = \sin 2t$, given $y(0) = 0, y'(0) = 0$

1 Take Laplace: $[s^2 Y(s) - 0 - 0] + 4Y(s) = \frac{2}{s^2 + 4}$

2 $Y(s)(s^2 + 4) = \frac{2}{s^2 + 4} \rightarrow Y(s) = \frac{2}{(s^2 + 4)^2}$

3 Inverse Laplace: $\mathcal{L}^{-1}\left\{\frac{2}{(s^2 + 4)^2}\right\} = \frac{1}{4}(\sin 2t - 2t \cos 2t)$

$$y = \frac{1}{4}(\sin 2t - 2t \cos 2t)$$

Module 5: Fourier Series

EULER'S FORMULAE — FULL RANGE $(-L, L)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right)$$

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx$$

HALF-RANGE SERIES AND PARSEVAL'S IDENTITY

Half-range cosine: $a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx, b_n = 0$

Half-range sine: $b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx, a_0 = a_n = 0$

Parseval's Identity: $\frac{1}{l} \int_{-l}^l [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$

Module 6: Fourier Transform

DEFINITIONS

Complex Fourier transform: $\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$

Inverse: $f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} d\omega$

Fourier sine transform: $F_s(\omega) = \int_0^{\infty} f(x) \sin \omega x dx$

Fourier cosine transform: $F_c(\omega) = \int_0^{\infty} f(x) \cos \omega x dx$

Parseval's: $\int_{-\infty}^{\infty} |f(x)|^2 dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{f}(\omega)|^2 d\omega$

Module 7: Z-Transform

DEFINITION AND STANDARD PAIRS

Definition: $Z\{x(n)\} = X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$

$$Z\{1\} = \frac{z}{z-1}$$

$$Z\{n\} = \frac{z}{(z-1)^2}$$

$$Z\{a^n\} = \frac{z}{z-a}$$

$$Z\{\cos n\theta\} = \frac{z(z - \cos \theta)}{z^2 - 2z \cos \theta + 1}$$

$$Z\{\sin n\theta\} = \frac{z \sin \theta}{z^2 - 2z \cos \theta + 1}$$

PROPERTIES AND INVERSE Z-TRANSFORM

Linearity: $Z\{ax(n) + by(n)\} = aX(z) + bY(z)$

Time shift: $Z\{x(n-k)\} = z^{-k} X(z)$

Convolution: $Z\{x * y\} = X(z) \cdot Y(z)$

Inverse (partial fractions): Expand $X(z)/z$ in partial fractions, then multiply by z and use table

— SOLVED — Z-TRANSFORM OF DIFFERENCE EQUATION

Solve $y(n+2) - 5y(n+1) + 6y(n) = 0$, $y(0) = 0$, $y(1) = 1$

1 Take Z-transform: $(z^2 - 5z + 6)Y(z) = z^2y(0) + z[y(1) - 5y(0)] = z$

2 $Y(z) = \frac{z}{(z-2)(z-3)} \rightarrow$ partial fractions: $\frac{-z}{z-2} + \frac{z}{z-3}$

3 Inverse Z: $Z^{-1}\left\{\frac{z}{z-a}\right\} = a^n$

$$y(n) = 3^n - 2^n$$

BMAT102L — Key Topics for Exam

- ☐ ODE: AE roots, CF+PI, variation of parameters
- ☐ PDE: Formation, Lagrange's method, separation of variables
- ☐ Laplace table, inverse via partial fractions, convolution
- ☐ Solving ODE/PDE with Laplace (Heaviside, impulse)
- ☐ Fourier series: full range, half range, Parseval's
- ☐ Z-transform: table, properties, inverse, difference equations

Complex Variables and Linear Algebra

Analytic Functions · Conformal Mapping · Complex Integration · Vector Spaces · Linear Algebra

Module 1: Analytic Functions

CAUCHY-RIEMANN EQUATIONS

$f(z) = u(x, y) + iv(x, y)$ is analytic if and only if:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Additionally, u and v must have continuous first-order partial derivatives

If analytic: $f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$

HARMONIC FUNCTIONS AND CONJUGATE

u is harmonic if $\nabla^2 u = u_{xx} + u_{yy} = 0$

If u is harmonic, find v (harmonic conjugate) using Milne-Thomson method

Milne-Thomson: $f'(z) = \phi_1(z, 0) - i\phi_2(z, 0)$ where $\phi_1 = u_x, \phi_2 = u_y$

— SOLVED — C-R EQUATIONS CHECK

Is $f(z) = z^2$ analytic? Find u, v .

1 $f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$, so $u = x^2 - y^2, v = 2xy$

2 $u_x = 2x = v_y = 2x \checkmark$ and $u_y = -2y = -v_x = -2y \checkmark$

C-R equations satisfied everywhere $\rightarrow f(z) = z^2$ is analytic everywhere (entire)

Module 2: Conformal Mapping

ELEMENTARY TRANSFORMATIONS

Translation: $w = z + c$ — shifts the plane

Rotation/Magnification: $w = az$ — rotates by $\arg(a)$ and scales by $|a|$

Inversion: $w = 1/z$ — maps circles to circles/lines

Square map: $w = z^2$ — doubles angles at origin

Exponential: $w = e^z$ — maps horizontal strips to sectors

BILINEAR (MÖBIUS) TRANSFORMATION

$$w = \frac{az + b}{cz + d}, \quad ad - bc \neq 0$$

Maps circles and lines to circles and lines

Cross-ratio preserved: $(w_1, w_2, w_3, w_4) = (z_1, z_2, z_3, z_4)$

Find transformation by mapping 3 points: use $\frac{(w - w_1)(w_2 - w_3)}{(w - w_3)(w_2 - w_1)} = \frac{(z - z_1)(z_2 - z_3)}{(z - z_3)(z_2 - z_1)}$

Module 3: Complex Integration

TAYLOR AND LAURENT SERIES

Taylor series (analytic at a): $f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z - a)^n$

Laurent series: $f(z) = \sum_{n=0}^{\infty} a_n (z - a)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z - a)^n}$

Residue = coefficient b_1 of $(z - a)^{-1}$ in Laurent series

SINGULARITIES AND RESIDUES

Removable singularity: Laurent series has no negative powers

Pole of order m : Laurent series has terms up to $(z - a)^{-m}$

Essential singularity: Infinitely many negative power terms

Residue at simple pole: $\text{Res}(f, a) = \lim_{z \rightarrow a} (z - a) f(z)$

Residue at pole of order m : $\frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} [(z - a)^m f(z)]$

CAUCHY'S THEOREMS

Cauchy-Goursat: $\oint_C f(z) dz = 0$ if f is analytic inside and on C

Cauchy's Integral Formula: $f^{(n)}(a) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - a)^{n+1}} dz$

Residue Theorem: $\oint_C f(z) dz = 2\pi i \sum_k \text{Res}(f, a_k)$

— SOLVED — RESIDUE THEOREM

Evaluate $\oint_{|z|=2} \frac{e^z}{z(z-1)} dz$

- 1 Poles inside $|z| = 2$: $z = 0$ and $z = 1$ (both simple poles)
- 2 $\text{Res}(f, 0) = \lim_{z \rightarrow 0} z \cdot \frac{e^z}{z(z-1)} = \frac{e^0}{0-1} = -1$
- 3 $\text{Res}(f, 1) = \lim_{z \rightarrow 1} (z-1) \cdot \frac{e^z}{z(z-1)} = \frac{e^1}{1} = e$
- 4 Result: $2\pi i(-1 + e) = 2\pi i(e - 1)$

$$\oint = 2\pi i(e - 1)$$

Modules 4–5: Vector Spaces and Linear Transformations

VECTOR SPACE DEFINITIONS

Subspace test: Closed under addition and scalar multiplication; contains zero vector

Linear independence: $c_1v_1 + c_2v_2 + \dots = 0$ implies all $c_i = 0$

Basis: Linearly independent spanning set

Dimension: Number of vectors in any basis

Rank-Nullity Theorem: $\text{rank}(A) + \text{nullity}(A) = n$ (number of columns)

LINEAR TRANSFORMATIONS

$T : V \rightarrow W$ is linear if $T(u + v) = T(u) + T(v)$ and $T(cu) = cT(u)$

Kernel: $\ker(T) = \{v : T(v) = 0\}$

Image: $\text{Im}(T) = \{T(v) : v \in V\}$

Rank: $\dim(\text{Im}(T))$; **Nullity:** $\dim(\ker T)$

Module 6: Inner Product Spaces

GRAM-SCHMIDT ORTHOGONALIZATION

Given basis $\{v_1, v_2, \dots, v_n\}$, construct orthonormal basis $\{e_1, e_2, \dots, e_n\}$:

$$u_1 = v_1; \quad e_1 = u_1 / \|u_1\|$$

$$u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1; \quad e_2 = u_2 / \|u_2\|$$

$$u_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j$$

Module 7: Eigenvalues and Cayley-Hamilton

EIGENVALUES AND EIGENVECTORS

Characteristic equation: $\det(A - \lambda I) = 0$

Eigenvector: $(A - \lambda_i I)\mathbf{x} = \mathbf{0}$

Properties: $\text{tr}(A) = \sum \lambda_i$; $\det(A) = \prod \lambda_i$

Cayley-Hamilton: Every matrix satisfies its own characteristic equation. To find A^{-1} or powers of A , use $p(\lambda) = 0 \Rightarrow p(A) = 0$

— SOLVED — EIGENVALUES AND CAYLEY-HAMILTON

For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find eigenvalues, eigenvectors, and verify Cayley-Hamilton.

1 Char eqn: $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0 \rightarrow \lambda = 1, 3$

2 For $\lambda = 1$: $(A - I)\mathbf{x} = \mathbf{0} \rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \mathbf{x} = \mathbf{0} \rightarrow \mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

3 For $\lambda = 3$: $\mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

4 Cayley-Hamilton: $A^2 - 4A + 3I = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} - \begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \checkmark$

$$\lambda_1 = 1, \mathbf{x}_1 = (1, -1)^T; \lambda_2 = 3, \mathbf{x}_2 = (1, 1)^T$$

GAUSSIAN ELIMINATION AND GAUSS-JORDAN

Gaussian elimination: Reduce augmented matrix $[A|b]$ to upper triangular form using row operations, then back-substitute

Gauss-Jordan: Continue until reduced row echelon form (identity matrix on left); solution reads directly

Consistency check: System has solution iff $\text{rank}(A) = \text{rank}([A|b])$

BMAT201L — Exam Priority Topics

- ☐ C-R equations, harmonic conjugate, Milne-Thomson method
- ☐ Bilinear transformation using cross-ratio
- ☐ Laurent series expansion and classification of singularities
- ☐ Residue theorem for contour integrals
- ☐ Rank-nullity theorem, basis, dimension
- ☐ Eigenvalues/vectors, Cayley-Hamilton theorem (find A^{-1})
- ☐ Gram-Schmidt process

Probability and Statistics

Random Variables · Distributions · Hypothesis Testing · ANOVA · Reliability

Module 1: Descriptive Statistics

MEASURES OF CENTRAL TENDENCY AND DISPERSION

Mean: $\bar{x} = \frac{1}{n} \sum x_i$; **Median:** middle value; **Mode:** most frequent

Variance: $\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$

Standard deviation: $\sigma = \sqrt{\sigma^2}$

Skewness (Pearson): $\gamma_1 = \frac{3(\text{Mean} - \text{Median})}{\sigma}$

Kurtosis: $\gamma_2 = \mu_4/\sigma^4$; Normal distribution has $\gamma_2 = 3$

Module 2: Random Variables

KEY FORMULAS

Discrete: $E(X) = \sum x P(X = x)$, $\text{Var}(X) = E(X^2) - [E(X)]^2$

Continuous: $E(X) = \int_{-\infty}^{\infty} x f(x) dx$

Covariance: $\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$

MGF: $M_X(t) = E(e^{tX})$; $E(X^n) = M_X^{(n)}(0)$

Joint distribution: $f_{XY}(x, y)$; marginal $f_X(x) = \int f_{XY}(x, y) dy$

Module 3: Correlation and Regression

CORRELATION COEFFICIENT AND REGRESSION LINES

Pearson's r: $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$; range: $-1 \leq r \leq 1$

Regression of Y on X: $y - \bar{y} = b_{yx}(x - \bar{x})$, where $b_{yx} = r \frac{\sigma_y}{\sigma_x}$

Regression of X on Y: $x - \bar{x} = b_{xy}(y - \bar{y})$, where $b_{xy} = r \frac{\sigma_x}{\sigma_y}$

Relationship: $r^2 = b_{yx} \cdot b_{xy}$

Rank correlation (Spearman): $\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$

Module 4: Probability Distributions

STANDARD DISCRETE DISTRIBUTIONS

Distribution	PMF	Mean	Variance
Binomial $B(n, p)$	$\binom{n}{x} p^x q^{n-x}$	np	npq
Poisson $P(\lambda)$	$e^{-\lambda} \lambda^x / x!$	λ	λ

STANDARD CONTINUOUS DISTRIBUTIONS

Distribution	PDF $f(x)$	Mean	Variance
Normal $N(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2
Exponential	$\lambda e^{-\lambda x}, x \geq 0$	$1/\lambda$	$1/\lambda^2$
Gamma $\Gamma(\alpha, \beta)$	$\frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}$	$\alpha\beta$	$\alpha\beta^2$
Weibull	$\frac{\beta}{\alpha} \left(\frac{x}{\alpha}\right)^{\beta-1} e^{-(x/\alpha)^\beta}$	$\alpha\Gamma(1 + 1/\beta)$	—

Standard Normal table (Z-scores)

For normal distribution problems, standardise: $Z = (X - \mu)/\sigma$. Then use Z-table. Common values:
 $P(Z \leq 1.645) = 0.95$; $P(Z \leq 1.96) = 0.975$; $P(Z \leq 2.576) = 0.995$

Modules 5–6: Hypothesis Testing

GENERAL PROCEDURE

- Step 1:** State H_0 (null) and H_1 (alternative)
- Step 2:** Choose significance level α (usually 0.05 or 0.01)
- Step 3:** Calculate test statistic
- Step 4:** Find critical value from table
- Step 5:** Decision: reject H_0 if test statistic falls in critical region

TEST STATISTICS

Test	Statistic	When to use
Z-test (mean)	$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$	Large sample ($n \geq 30$), σ known
t-test (mean)	$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}, \text{ df} = n - 1$	Small sample, σ unknown
F-test (variance)	$F = s_1^2/s_2^2$	Comparing two variances
χ^2 (goodness of fit)	$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$	Categorical data, independence

Module 6: ANOVA and Design of Experiments

ONE-WAY ANOVA

$SST = SSB + SSW$ (Total = Between groups + Within groups)

$MSB = SSB/(k - 1)$; $MSW = SSW/(N - k)$

F-statistic: $F = MSB/MSW$

Reject H_0 (all means equal) if $F > F_{\alpha, k-1, N-k}$

CRD vs RBD vs LSD

CRD (Completely Randomised): Treatments assigned randomly, one-way ANOVA

RBD (Randomised Block): Controls one extraneous variable (blocks), two-way ANOVA

LSD (Latin Square): Controls two extraneous variables simultaneously, three-way ANOVA

Module 7: Reliability Engineering

RELIABILITY FUNCTIONS

Reliability: $R(t) = P(T > t) = 1 - F(t) = e^{-\lambda t}$ (exponential)

Hazard function: $h(t) = f(t)/R(t) = \lambda$ (constant for exponential)

MTTF: $= \int_0^\infty R(t) dt = 1/\lambda$

Series system: $R_s = R_1 \cdot R_2 \cdots R_n$

Parallel system: $R_s = 1 - (1 - R_1)(1 - R_2) \cdots (1 - R_n)$

BMAT202L — Exam Priority Topics

- ☐ Binomial and Poisson distribution problems
- ☐ Normal distribution — standardisation, Z-table usage
- ☐ Correlation coefficient and regression lines
- ☐ Z-test, t-test, F-test, chi-square — know which to apply when
- ☐ ANOVA: $SST = SSB + SSW$, MSB, MSW, F-ratio calculation
- ☐ Series and parallel reliability, MTTF

Master Formula Quick Reference

One-page revision sheet — all four courses

BMAT101L — Calculus

$$\nabla f = \partial_x \hat{i} + \partial_y \hat{j} + \partial_z \hat{k}$$

$$D = f_{xx}f_{yy} - (f_{xy})^2 \text{ for saddle/max/min}$$

$$B(m, n) = \Gamma(m)\Gamma(n)/\Gamma(m+n)$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} \text{ (Stokes)}$$

$$\oiint \mathbf{F} \cdot d\mathbf{S} = \iiint \nabla \cdot \mathbf{F} dV \text{ (Gauss)}$$

BMAT201L — Complex Variables & LA

$$\text{C-R: } u_x = v_y, u_y = -v_x$$

$$\text{Residue theorem: } \oint = 2\pi i \sum \text{Res}$$

$$\text{Bilinear: } w = (az + b)/(cz + d)$$

$$\text{Rank-Nullity: rank} + \text{nullity} = n$$

$$\det(A) = \prod \lambda_i, \text{tr}(A) = \sum \lambda_i$$

BMAT102L — DE & Transforms

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a) \text{ (1st shift)}$$

$$\mathcal{L}\{f''\} = s^2F - sf(0) - f'(0)$$

$$\text{Fourier: } a_n = \frac{1}{l} \int_{-l}^l f \cos \frac{n\pi x}{l} dx$$

$$Z\{a^n\} = z/(z-a)$$

$$Z\{n\} = z/(z-1)^2$$

BMAT202L — Probability & Statistics

$$\text{Binomial: } E = np, \text{Var} = npq$$

$$\text{Poisson: } P(X=x) = e^{-\lambda} \lambda^x / x!$$

$$r = \text{Cov}(X, Y) / (\sigma_X \sigma_Y)$$

$$F = \text{MSB}/\text{MSW} \text{ (ANOVA)}$$

$$\text{Series: } R_s = \prod R_i; \text{Parallel: } R_s = 1 - \prod (1 - R_i)$$

Good luck in your exams.

Understand the concept. Practice the formula. Attempt the example.

That's the only revision strategy that works.